

波束成形

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Info

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内容	均匀线阵，均匀面阵，均匀圆阵，均匀圆环阵原理分析与仿真

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均匀圆阵 UCA

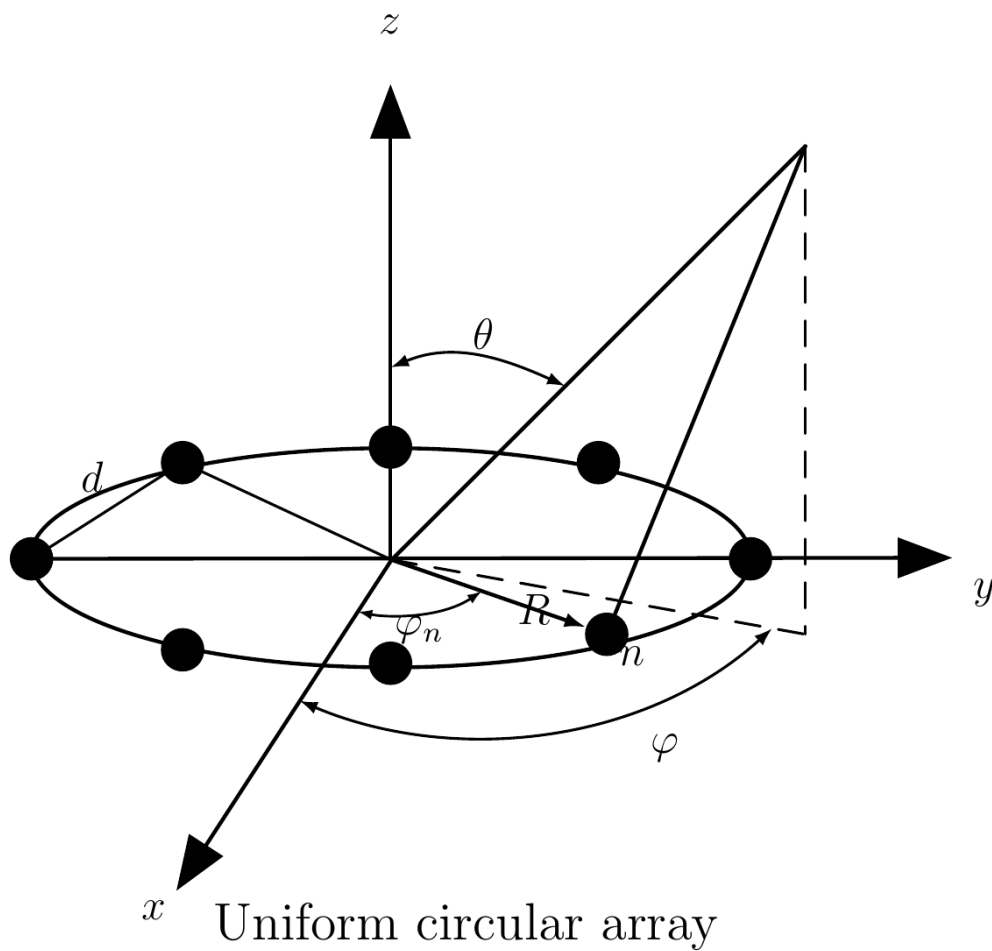


图 1: 均匀圆阵的几何结构示意图

公式推导:

Tips: θ : the elevation plane , φ : the azimuth plane

$$\varphi_n = \frac{2\pi n}{N}, \quad n = 0, 1, \dots, N-1$$

$$\mathbf{r}_n = (R \cdot \cos \varphi_n, R \cdot \sin \varphi_n, 0)$$

$$\mathbf{u}(\varphi, \theta) = (\cos \varphi \cdot \sin \theta, \sin \varphi \cdot \sin \theta, \cos \theta)$$

阵元在观察方向上的投影:

$$\begin{aligned}
\mathbf{r}_n \cdot \mathbf{u} &= R \cos \varphi_n \cdot \cos \varphi \sin \theta + R \sin \varphi_n \cdot \sin \varphi \sin \theta + 0 \cdot \cos \theta \\
&= R \sin \theta \cdot (\cos \varphi_n \cos \varphi + \sin \varphi_n \sin \varphi) \\
&= R \sin \theta \cdot \cos(\varphi - \varphi_n)
\end{aligned}$$

即阵元 $\mathbf{n}$ 与圆心的波程差。

The array factor:

$$F(\varphi, \theta) = \sum_{n=0}^{N-1} A_n \cdot e^{j \cdot [\alpha_n - k \cdot R \cdot \sin \theta \cdot \cos(\varphi - \varphi_n)]}$$

其中:

$$\alpha_n = k \cdot R \cdot \sin \theta_0 \cdot \cos(\varphi_0 - \varphi_n), k = \frac{2\pi}{\lambda}$$

仿真文件:

uca_3d_pattern.m

均匀圆环阵 UCCA

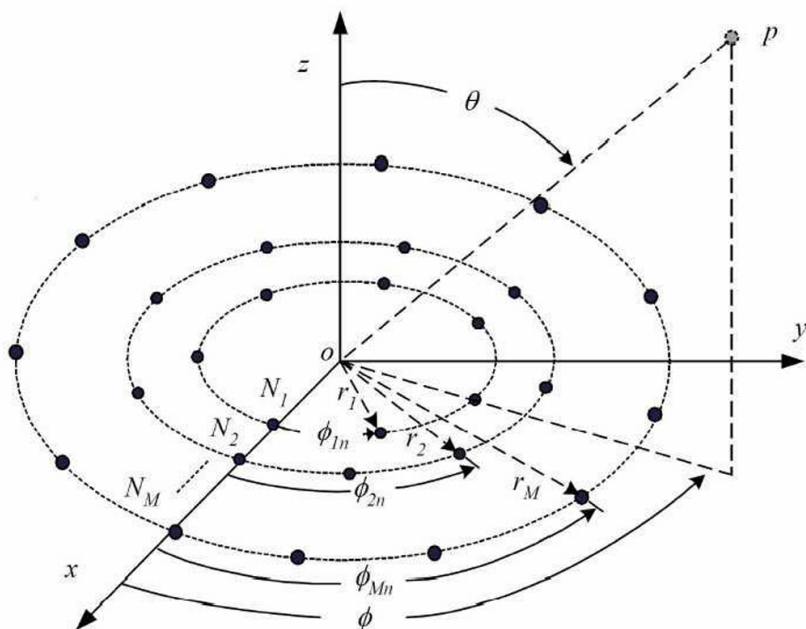


图 2: 均匀圆环阵的几何结构示意图

公式推导:

$$F(\varphi, \theta) = 1 + \sum_{i=1}^M \sum_{j=1}^{N_i} A_{ij} \cdot e^{j[\alpha_{ij} - k \cdot r_i \cdot \sin \theta \cos(\varphi - \varphi_{ij})]}$$

其中:

$$\alpha_{ij} = k \cdot r_i \cdot \sin \theta_0 \cdot \cos(\varphi_0 - \varphi_{ij})$$

$$\varphi_{ij} = \frac{2\pi(j-1)}{N_i}, j = 1, 2, \dots, N_i;$$

仿真文件:

ucca_3d_pattern.m

均匀线阵 ULA

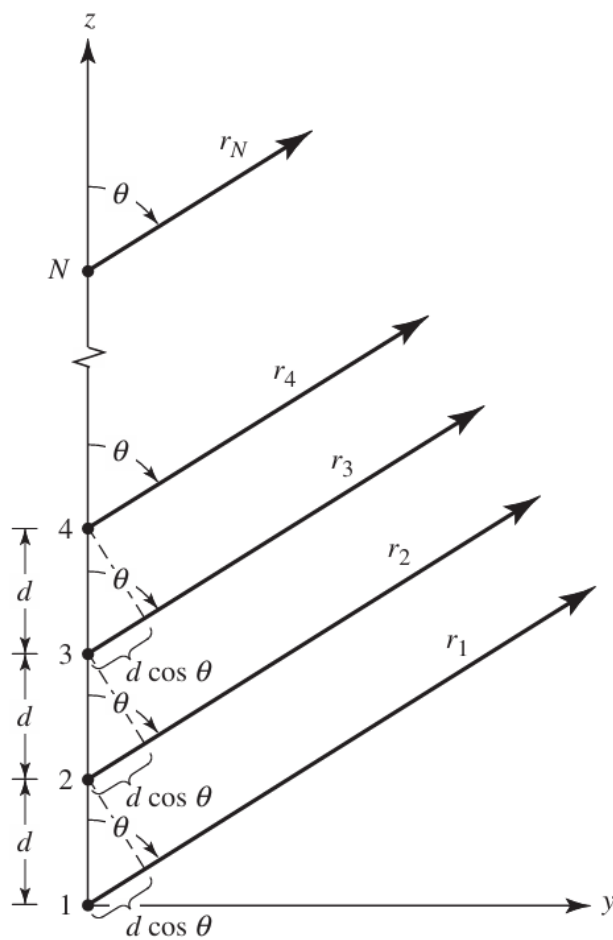


图 3: 均匀线阵的几何结构示意图

公式推导：

相邻两阵元之间的相位差：the array phase function

$$\psi = kd \cos \theta + \alpha$$

The array factor (所有阵元振幅相同为 1):

$$\begin{aligned} AF &= \sum_{n=0}^{N-1} e^{jn(k \cdot d \cdot \cos \theta + \alpha)} = \sum_{n=0}^{N-1} e^{jn\psi} \\ &\stackrel{\text{等比数列求和}}{=} \frac{e^{jN\psi} - 1}{e^{j\psi} - 1} \\ &= \frac{e^{j\frac{N}{2}\psi} (e^{j\frac{N}{2}\psi} - e^{-j\frac{N}{2}\psi})}{e^{j\frac{\psi}{2}} (e^{j\frac{\psi}{2}} - e^{-j\frac{\psi}{2}})} \\ &= e^{j(N-1)\frac{\psi}{2}} \cdot \frac{\sin(\frac{N\psi}{2})}{\sin(\frac{\psi}{2})} \end{aligned}$$

The array factor after phase shift and normalization becomes:

$$AF = \frac{1}{N} \cdot \frac{\sin(\frac{N\psi}{2})}{\sin(\frac{\psi}{2})}$$

仿真文件：

ula_polar_pattern.m

均匀面阵 UPA

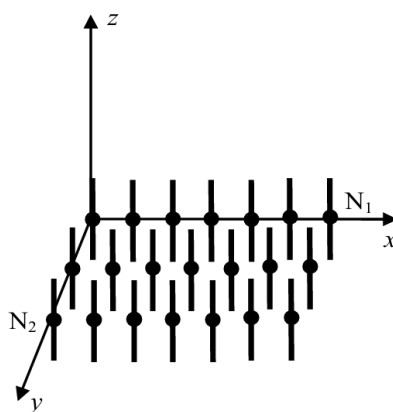


Fig. 2 The planar array antennas

图 4: 均匀二维面阵的几何结构示意图

公式推导:

$$\mathbf{r}_{m,n} = (md_x, nd_y, 0)$$

$$\mathbf{u}(\varphi, \theta) = (\cos \varphi \cdot \sin \theta, \sin \varphi \cdot \sin \theta, \cos \theta)$$

阵元 (m,n) 与原点的波程差:

$$\mathbf{r}_{m,n} \cdot \mathbf{u} = md_x \cdot \cos \varphi \sin \theta + nd_y \cdot \sin \varphi \sin \theta + 0 \cdot \cos \theta$$

阵元 (m,n) 的几何相位 (原点相位为 0):

$$m \cdot (k \cdot d_x \cos \varphi \sin \theta) + n \cdot (k \cdot d_y \sin \varphi \sin \theta)$$

令:

$$\psi_x = kd_x \cos \varphi \sin \theta + \alpha_x$$

$$\psi_y = kd_y \sin \varphi \sin \theta + \alpha_y$$

其中:

$$\alpha_x = -kd_x \sin \theta_0 \cos \phi_0$$

$$\alpha_y = -kd_y \sin \theta_0 \sin \phi_0$$

The array factor (所有阵元振幅相同为 1):

$$\begin{aligned} AF &= \sum_{m=0}^{N_1-1} \sum_{n=0}^{N_2-1} e^{j(m\psi_x + n\psi_y)} = \sum_{m=0}^{N_1-1} \sum_{n=0}^{N_2-1} e^{jm\psi_x} \cdot e^{jn\psi_y} \\ &= \left(\sum_{m=0}^{N_1-1} e^{jm\psi_x} \right) \left(\sum_{n=0}^{N_2-1} e^{jn\psi_y} \right) = AF_x \cdot AF_y \\ &= \frac{\sin \left(\frac{N_1\psi_x}{2} \right)}{N_1 \sin \left(\frac{\psi_x}{2} \right)} \cdot \frac{\sin \left(\frac{N_2\psi_y}{2} \right)}{N_2 \sin \left(\frac{\psi_y}{2} \right)} \end{aligned}$$

仿真文件:

upa_3d_pattern.m

3 维阵列:

同理可得:

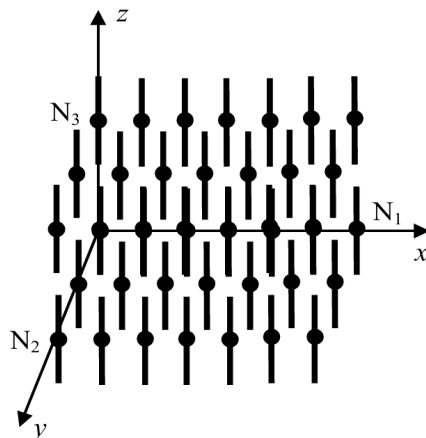


Fig. 3 Three – dimensional array antennas

图 5: 均匀三维面阵的几何结构示意图

$$\psi_x = kd_x \sin \theta \cos \varphi + \alpha_x$$

$$\psi_y = kd_y \sin \theta \sin \varphi + \alpha_y$$

$$\psi_z = kd_z \cos \theta + \alpha_z$$

其中:

$$\alpha_x = -kd_x \sin \theta_0 \cos \phi_0$$

$$\alpha_y = -kd_y \sin \theta_0 \sin \phi_0$$

$$\alpha_z = -kd_z \cos \theta_0$$

The array factor (所有阵元振幅相同为 1):

$$AF = AF_x \cdot AF_y \cdot AF_z = \frac{\sin \left(\frac{N_1 \psi_x}{2} \right)}{N_1 \sin \left(\frac{\psi_x}{2} \right)} \cdot \frac{\sin \left(\frac{N_2 \psi_y}{2} \right)}{N_2 \sin \left(\frac{\psi_y}{2} \right)} \cdot \frac{\sin \left(\frac{N_3 \psi_z}{2} \right)}{N_3 \sin \left(\frac{\psi_z}{2} \right)}$$

仿真文件:

uva_3d_pattern.m

半波偶极子的归一化方向图函数:

$$F(\theta) = \frac{\cos \left(\frac{\pi}{2} \cos \theta \right)}{\sin \theta}$$

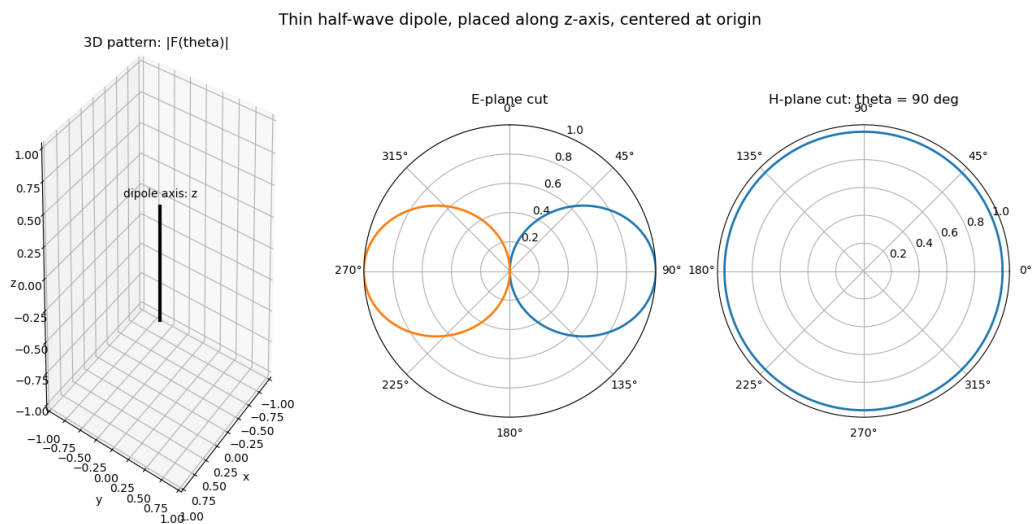


图 6: 半波偶极子的归一化方向图函数